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Experiment: 7

Analog Beamforming Analysis Using MATLAB

OBJECTIVE 1 :

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Beamforming Gain (dB)
```

```
figure
```

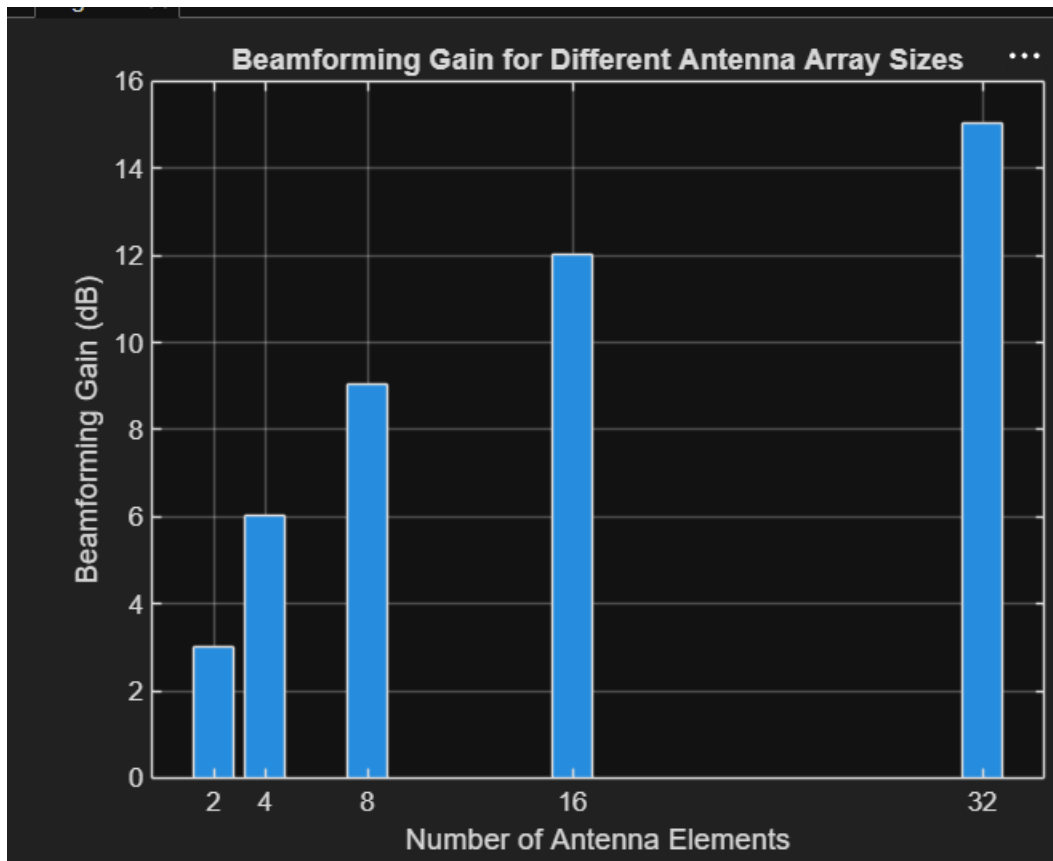
```
bar(antennas, gain)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

```
title('Beamforming Gain for Different Antenna Array Sizes')
```



OBJECTIVE 2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width  
(degrees)
```

```
figure
```

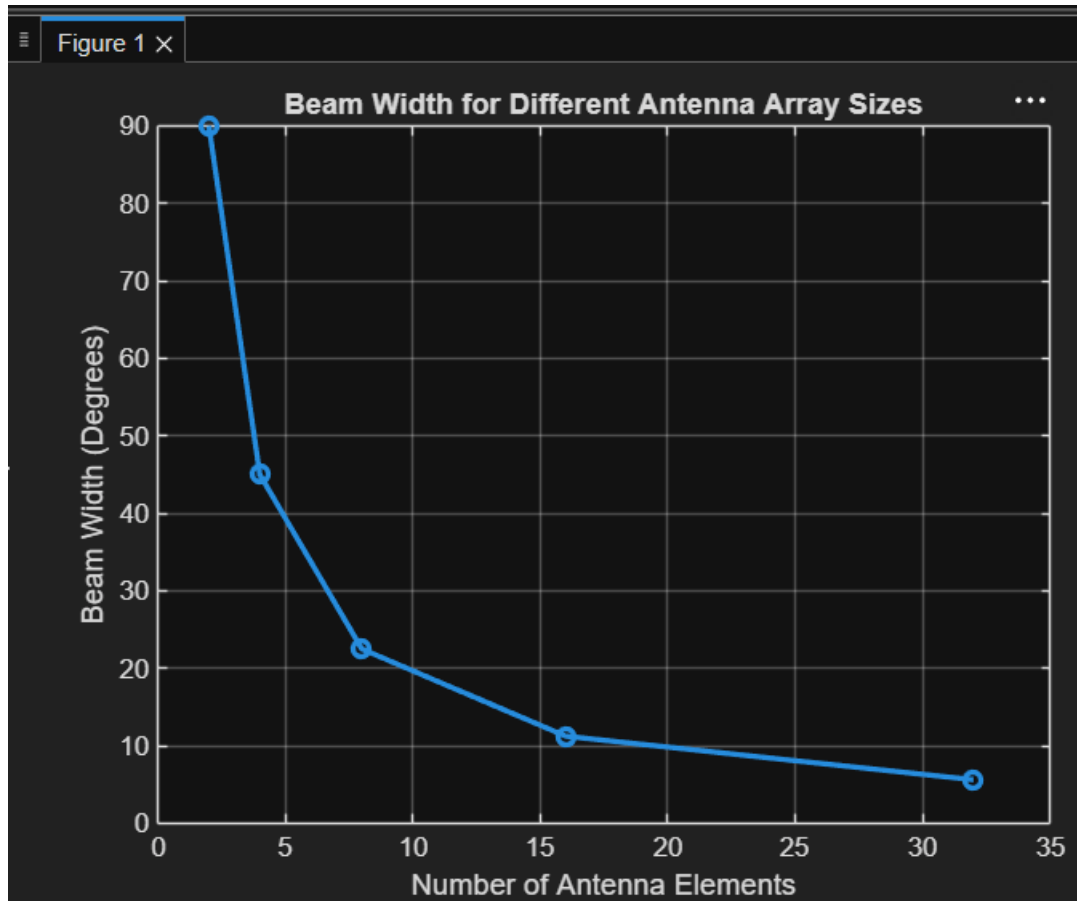
```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```



OBJECTIVE 3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
```

```
gain = [1 1 1 1 1]; % Constant normalized gain
```

```
figure
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```

